Weekly Quiz 1

Student Edition

Sets, diagrams, and counting

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. Express "A or B, but not both" using unions, intersections, and complements. **[3 points]**

2. A fair coin is tossed three times. Find the probability of exactly two heads. **[3 points]**

3. In a group of 70 students, 38 study French, 27 study Spanish, and 12 study both. How many study neither language? **[4 points]**

Weekly Quiz 2

Student Edition

Probability spaces and events

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. Two fair dice are rolled. Find the probability that the maximum is 4. **[4 points]**

2. Find the probability of at least one six in four independent rolls of a fair die. **[3 points]**

3. Derive $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ by splitting events into disjoint pieces. **[3 points]**

Weekly Quiz 3

Student Edition

Conditional probability and Bayes' rule

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. An urn contains 3 red and 2 blue balls. Two are drawn without replacement. Find the probability that the second ball is red. **[3 points]**

2. A condition has prevalence 2%. A test has sensitivity 90% and specificity 95%. Find the probability that a person who tests positive has the condition. **[4 points]**

3. Suppose $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.2$. Are A and B independent? Find $\mathbb{P}(A \mid B)$. **[3 points]**

Weekly Quiz 4

Student Edition

Random variables, PMFs, densities, and CDFs

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. A random variable X takes values $0, 1, 2$ with probabilities $0.2, 0.5, 0.3$. Write its CDF $F(x) = \mathbb{P}(X \le x)$ piecewise. **[4 points]**

2. Let $f(x) = cx$ for $0 \le x \le 2$ and $f(x) = 0$ otherwise. Find c and $\mathbb{P}(X > 1)$. **[4 points]**

3. If X has a continuous distribution, what is $\mathbb{P}(X = 1)$? Why? **[2 points]**

Weekly Quiz 5

Student Edition

Expectation, variance, and moments

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. For X with probabilities $0.2, 0.5, 0.3$ at $0, 1, 2$, compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$. **[4 points]**

2. If $X \sim \mathrm{Bin}(20, 0.3)$, find its mean and variance. Explain briefly using indicators. **[3 points]**

3. Given $\mathbb{E}[X] = 4$ and $\mathbb{E}[Y] = -1$, find $\mathbb{E}[3X - 2Y + 5]$. Is independence needed? **[3 points]**

Weekly Quiz 6

Student Edition

Standard probability distributions

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. If $X \sim \text{Bin}(5, 0.4)$, find $\mathbb{P}(X = 2)$. **[3 points]**

2. A waiting time is exponential with mean 4 minutes. Find the probability that the wait exceeds 6 minutes. **[3 points]**

3. Suppose $X \sim N(100, 15^2)$. Express $\mathbb{P}(X \leq 130)$ using the standard normal CDF Φ, and evaluate using $\Phi(2) \approx 0.9772$. **[4 points]**

Weekly Quiz 7

Student Edition

Joint, marginal, and conditional distributions

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

For the next three questions, use the joint probability table

	$Y = 0$	$Y = 1$	$\mathbb{P}(X = x)$
$X = 0$	0.10	0.20	
$X = 1$	0.30	0.40	
$\mathbb{P}(Y = y)$			1

1. Find both marginal distributions. **[3 points]**

2. Find $\mathbb{P}(X = 1 \mid Y = 0)$ and decide whether X and Y are independent. **[3 points]**

3. Compute $\text{Cov}(X, Y)$. **[4 points]**

Weekly Quiz 8

Student Edition

Sampling distributions, the LLN, and the CLT

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. A population has standard deviation 12. For samples of size 64, find the standard error of the sample mean. **[3 points]**

2. In one sentence each, distinguish the law of large numbers from the central limit theorem. **[3 points]**

3. If 100 independent Bernoulli trials have success probability 0.4, give the approximate mean and standard deviation of the sample proportion. Standardize the event $\hat{p} \leq 0.50$. **[4 points]**

Weekly Quiz 9

Student Edition

Estimation, confidence intervals, and tests

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. In a random sample, 330 of 600 respondents answer yes. Compute the usual approximate 95% confidence interval for the population proportion, using $z^* = 1.96$. **[4 points]**

2. Give a correct repeated-sampling interpretation of a 95% confidence procedure. **[3 points]**

3. A valid test returns $p = 0.03$. State an appropriate conclusion at level $\alpha = 0.05$, and name one conclusion the p-value does *not* justify. **[3 points]**

Weekly Quiz 10

Student Edition

Bayesian inference and prediction

20 minutes • 10 points • Show the essential steps.

Name: _____ **Date:** _____

1. A Bernoulli success probability has prior $p \sim \text{Beta}(2, 2)$. After observing 8 successes and 2 failures, find the posterior distribution. **[4 points]**

2. Find the posterior mean and the posterior predictive probability that the next trial is a success. **[3 points]**

3. Contrast a 95% Bayesian credible interval with a 95% frequentist confidence interval in one or two sentences. **[3 points]**